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C:\Users\HP>HP>OneDrive>Desktop>python projects>learn>SHAI.py
1 import numpy as np
2
3 def init_pop(pop_size):
4     return np.random.randint(8, size=(pop_size, 8))
5
6 def calc_fitness(population):
7     fitness_vals = []
8     for x in population:
9         penalty = 0
10        for i in range(8):
11            r = x[i]
12            for j in range(8):
13                if i == j:
14                    continue
15                d = abs(i - j)
16                if x[j] in [r, r - d, r + d]:
17                    penalty += 1
18            fitness_vals.append(penalty)
19    return -1 * np.array(fitness_vals)
20
21 def selection(population, fitness_vals):
22    probs = fitness_vals.copy()
23    probs += abs(probs.min()) + 1
24    probs = probs / probs.sum()
25    n = len(population)
26    indices = np.arange(n)
27    selected_indices = np.random.choice(indices, size=n, p=probs)
28    selected_population = population[selected_indices]
29    return selected_population
30
31 def crossover(parent1, parent2, pc):
32    r = np.random.random()
33    if r < pc:
34        m = np.random.randint(1, 8)
35        child1 = np.concatenate([parent1[:m], parent2[m:]])
36        child2 = np.concatenate([parent2[:m], parent1[m:]])
37    else:
38        child1 = parent1.copy()
39        child2 = parent2.copy()
40    return child1, child2
41
42 def mutation(individual, pm):
43    r = np.random.random()
44    if r < pm:
45        m = np.random.randint(8)
46        individual[m] = np.random.randint(8)
47    return individual
48
49 def crossover_mutation(selected_pop, pc, pm):
50    n = len(selected_pop)
51    new_pop = np.empty((n, 8), dtype=int)
52    for i in range(0, n, 2):
53        parent1 = selected_pop[i]
54        parent2 = selected_pop[i + 1]
55        child1, child2 = crossover(parent1, parent2, pc)
56        new_pop[i] = child1
57        new_pop[i + 1] = child2
58    for i in range(n):
59        mutation(new_pop[i], pm)
60    return new_pop
61
62 def eight_queens(pop_size, max_generation, pc=0.7, pm=0.01):
63    population = init_pop(pop_size)
64    best_fitness_overall = None
65    for i gen in range(max_generation):
66        fitness_vals = calc_fitness(population)
67        best_i = fitness_vals.argmax()
68        best_fitness = fitness_vals[best_i]
69        if best_fitness_overall is None or best_fitness > best_fitness_overall:
70            best_fitness_overall = best_fitness
71            best_solution = population[best_i]
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37 else:
38     child1 = parent1.copy()
39     child2 = parent2.copy()
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69         if best_fitness_overall is None or best_fitness > best_fitness_overall:
70             best_fitness_overall = best_fitness
71             best_solution = population[best_i]
72             print(f'\ri_gen={i_gen:06}-f={-best_fitness_overall:03}', end='')
73             if best_fitness == 0:
74                 print('\nfound optimal solution')
75                 break
76         selected_pop = selection(population, fitness_vals)
77         population = crossover_mutation(selected_pop, pc, pm)
78     print()
79     print(best_solution)
80     return best_solution
81
82 eight_queens(pop_size=100, max_generation=10000, pc=0.7, pm=0.01)
```

Snipping Tool

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python + -

PS C:\Users\HP> & C:\Users\HP\AppData\Local\Programs\Python\Python312\python.exe "c:\Users\HP\OneDrive\Desktop\python projects\learn\SHAI.py"
i_gen=000077-f=002

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